Isotope effect, Cooper pairs and BCS theory

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1 Phonons and isotope mass

The variation in phonon frequency with the inverse square root of isotope mass, $\omega_{phonon} \propto \frac{1}{\sqrt{M}}$, arises from the physics of lattice vibrations in a crystal, where phonons are quantized vibrational modes. This relationship emerges naturally when considering the dynamics of atoms in a crystal lattice, modeled as a system of masses connected by springs (the harmonic approximation). Let's derive this step by step.

1.1 Understanding Phonons and Lattice Vibrations

Phonons are quantized collective vibrational modes of atoms in a crystal lattice. In a solid, atoms are arranged in a periodic lattice and interact with their neighbours via interatomic forces, often called harmonic (spring-like). The frequency of these vibrations depends on the mass of the atoms and the strength of the interatomic interactions. When we consider the effect of isotopic substitution, we're changing the mass (M) of the atoms in the lattice (e.g., replacing ^{12}C with ^{13}C in a carbon-based material like diamond). Isotopes have the same electronic structure, so the interatomic forces (and thus the "spring constant" of the bonds) remain unchanged, but the mass of the atoms changes.

1.2 Model the Lattice as a Harmonic Oscillator

To understand phonon frequencies, start with a simple model: a one-dimensional chain of atoms. Imagine a chain of atoms with mass M, connected by springs with spring constant k. This spring constant k represents the strength of the interatomic forces, which depends on the electronic structure and is unaffected by isotopic mass. For a single harmonic oscillator (one atom vibrating against a fixed point), the frequency of oscillation is given by:

$$\omega_{phonon} = \sqrt{\frac{k}{M}}$$

$$T_C M^{1/2} = constant.$$

This is the classical frequency of a mass M on a spring with spring constant k. Notice the $\frac{1}{\sqrt{M}}$ dependence: as the mass increases, the frequency decreases because a heavier mass oscillates more slowly for the same restoring force.

1.3 Extending to a Crystal Lattice

In a crystal lattice, the atoms are coupled, and the vibrations are collective modes (phonons). Consider a one-dimensional lattice with atoms of mass M, lattice spacing a, and nearest-neighbor interactions modeled by a spring constant k. The equation of motion for the displacement u_n of the n-th atom is:

$$M\frac{d^2u_n}{dt^2} = k(u_{n+1} - u_n) + k(u_{n-1} - u_n) = k(u_{n+1} + u_{n-1} - 2u_n)$$

Assume a traveling wave solution for the displacement, typical for phonons:

$$u_n = u_0 e^{i(qna-\omega t)}$$

where q is the wavevector, ω is the phonon frequency, and a is the lattice spacing. Substitute into the equation of motion:

$$-M\omega^2 u_0 e^{i(qna-\omega t)} = k\left(e^{iqa} + e^{-iqa} - 2\right) u_0 e^{i(qna-\omega t)}$$

Cancel the common exponential terms:

$$-M\omega^2 = k\left(e^{iqa} + e^{-iqa} - 2\right) = k(2\cos(qa) - 2) = k \cdot 2(\cos(qa) - 1)$$

$$M\omega^2 = 2k(1 - \cos(qa)) = 4k\sin^2\left(\frac{qa}{2}\right)$$

$$\omega = \sqrt{\frac{4k}{M}}\left|\sin\left(\frac{qa}{2}\right)\right|$$

This is the dispersion relation for phonons in a 1D lattice. The key point is the $\omega \propto \frac{1}{\sqrt{M}}$ dependence, which arises from the $\sqrt{\frac{k}{M}}$ term.

1.4 Application to Isotopes

When the mass M changes due to isotopic substitution (e.g., from M_1 to M_2), the spring constant k remains the same because the electronic interactions (bond strength) are unchanged—isotopes differ only in nuclear mass, not in chemical properties. Thus, the phonon frequency scales as:

$$\omega \propto \sqrt{\frac{k}{M}} \propto \frac{1}{\sqrt{M}}$$

For two isotopes with masses M_1 and M_2, the ratio of their phonon frequencies is:

$$\frac{\omega_2}{\omega_1} = \sqrt{\frac{M_1}{M_2}}$$

If $M_2 > M_1$, then $\omega_2 < \omega_1$, meaning the heavier isotope has a lower phonon frequency.

1.5 Physical Interpretation of phonons

The $\frac{1}{\sqrt{M}}$ dependence makes sense physically: heavier atoms oscillate more slowly because their inertia is greater, but the restoring force (from the interatomic potential, i.e., k) remains the same. In a 3D crystal, the same principle applies. The phonon dispersion relation depends on the direction and branch (acoustic or optical), but the frequency always scales with $\frac{1}{\sqrt{M}}$ when considering the effect of mass.

1.6 Optical Phonons and Diatomic Lattices

For a diatomic lattice (e.g., with two different masses M_1 and M_2), the optical phonon frequency at the zone center ($q = 0$) also shows this dependence. The optical mode involves the two atoms vibrating out of phase, and the frequency is determined by the reduced mass $\mu = \frac{M_1 M_2}{M_1 + M_2}$. If one mass is fixed and the other changes due to an isotope, the frequency still scales as $\frac{1}{\sqrt{\mu}}$, which reduces to $\frac{1}{\sqrt{M}}$ for the varying isotope.

Conclusion

The phonon frequency varies as $\frac{1}{\sqrt{M}}$ because the frequency of lattice vibrations depends on the inverse square root of the atomic mass in a harmonic system, while the interatomic force constants remain unchanged with isotopic substitution. This is a direct consequence of the dynamics of coupled harmonic oscillators in the lattice, as seen in both simple 1D models and more complex 3D crystals.

2 Isotope effect in superconductors

The Isotope Effect in Superconductivity The isotope effect refers to the observed dependence of a superconductor's critical temperature (T_c) on the isotopic mass of its constituent atoms. Discovered experimentally in the 1950s, this phenomenon provided critical evidence for the role of lattice vibrations (phonons) in superconductivity.

Theoretical Basis In a conventional superconductor, the transition to the superconducting state occurs when electrons form pairs that move without resistance. The isotope effect arises because the pairing mechanism involves interactions mediated by phonons, which are quantized lattice vibrations. The frequency of these phonons (ω) depends on the ionic mass (M) of the lattice atoms. In a simple harmonic oscillator model, the phonon frequency scales as:

$$\omega \propto \frac{1}{\sqrt{M}}.$$

The critical temperature T_c is related to the strength of the electron-phonon interaction. In BCS theory, T_c is approximately given by:

$$T_c \propto \hbar\omega e^{-1/(N(0)V)},$$

where $\hbar$ is the reduced Planck constant, $N(0)$ is the electronic density of states at the Fermi level, and V is the effective electron-phonon coupling strength. If V is independent of isotopic mass, then:

$$T_c \propto \omega \propto M^{-1/2}.$$

Thus, T_c varies inversely with the square root of the isotopic mass, leading to the relation:

$$T_c M^{1/2} = \text{constant}.$$

Experimental Evidence The isotope effect was first confirmed in mercury by measuring T_c for different isotopes (e.g., ^{200}Hg and ^{204}Hg). Experiments showed that $T_c \propto M^{-\alpha}$, with $\alpha \approx 0.5$ for many elemental superconductors, consistent with the phonon-mediated pairing hypothesis. Deviations from $\alpha = 0.5$ occur in some materials due to variations in $N(0)$ or V, but the effect strongly supports the idea that lattice dynamics are integral to superconductivity.

Significance The isotope effect provided a key clue that superconductivity is not purely an electronic phenomenon but involves the lattice. This insight guided the development of BCS theory, which explicitly incorporates phonon-mediated electron pairing.

3 Reference system: System of degenerate states

Consider a system of four states $|1\rangle, |2\rangle, |3\rangle, |4\rangle$ forming a basis in which the Hamiltonian 0 is diagonal, with four identical eigenvalues ϵ. We then add an interaction potential $\hat{V}$ that has matrix elements on this basis that are all identical and negative, with value V. $\langle j|\hat{V}|i\rangle = V_{ji} = V$. Finding the new energy levels, i.e. the new eigenvalues of the Hamiltonian, involves solving the system of equations

$$\begin{pmatrix} \epsilon - V - E & V & V & V \\ V & \epsilon - V - E & V & V \\ V & V & \epsilon - V - E & V \\ V & V & V & \epsilon - V - E \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

The solutions are the values of Emaking the determinant vanish or,

$$(E - \epsilon)^3 (E - \epsilon + 4V) = 0$$

which leads to three levels of energy equal to the initial energy and a fourth lower energy level, $E_1 = \epsilon$, $E_2 = \epsilon$, $E_3 = \epsilon$; $E_\alpha = \epsilon - 4V$. The eigenvector $|\psi\rangle$ associated with this last level is the linear symmetric combination of the states $|1\rangle, |2\rangle, |3\rangle$ and $|4\rangle$,

$$|\psi\rangle = \frac{1}{4}(|1\rangle + |2\rangle + |3\rangle + |4\rangle).$$

In the language of quantum mechanics $|\psi\rangle$ is a stationary state, which means that:

- if the particle is in this state at time $t = 0$ it will stay in this state indefinitely, with quantum phase factor $e^{-iEt/\hbar}$;
- if at time $t = 0$, the particle is, for example, in the state $|1\rangle$, which is not an eigenvector of the complete Hamiltonian, at later times, it will be found with non-zero probabilities in the other states $|2\rangle, |3\rangle$ and $|4\rangle$. The time-dependent equation gives the evolution of the system

$$i\frac{d}{dt}c_i(t) = \sum_j V_{ij}c_j(t)$$

  where c_i is the probability amplitude of finding the particle in the state $|i\rangle$.

4 System of non-degenerate states

System with non-degenerate states

We now consider the case where eigenvectors of the non-interacting system $|1\rangle, |2\rangle, |3\rangle$ and $|4\rangle$ correspond to different energies $\epsilon_1, \epsilon_2, \epsilon_3$ and ϵ_4. Introduction of the potential $\hat{V}$ leads to the eigenvalue equation

$$\begin{pmatrix} \epsilon_1 - V - E & -V & -V & -V \\ -V & \epsilon_2 - V - E & -V & -V \\ -V & -V & \epsilon_3 - V - E & -V \\ -V & -V & -V & \epsilon_4 - V - E \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

i.e., to four equations with four unknowns

$$V(c_1 + c_2 + c_3 + c_4) = c_i(\epsilon_i - E); \quad i \in [1, 4]$$

or, equivalently, to the four relations

$$\frac{c_i}{V} = \frac{c_1 + c_2 + c_3 + c_4}{\epsilon_i - E}; \quad i \in [1, 4].$$

Summing each of the two sides over the index "i" we then obtain a general relation, and cancelling $\sum_{i=1}^4 c_i$:

$$\frac{1}{V} = \sum_{i=1}^{4} \frac{1}{\epsilon_i - E}.$$

If $\epsilon_1 = \epsilon_2 = \epsilon_3 = \epsilon_4 = \epsilon$, we recover the solution $E = \epsilon - NV$, where N is the order of the matrix.

When the energies ϵ_i differ ($\epsilon_1 < \epsilon_2 < \epsilon_3 < \epsilon_4$), the equation always has one, and only one, energy lower than the lowest of the original energies as appears clearly by examination of the energy levels. It is the only solution for which all the c_i are positive.

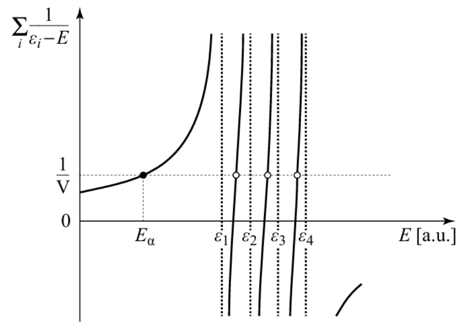

Figure 1: Bound state in an initially non-degenerate state An interaction that adds a term $-V$ to each element of an initially non-degenerate Hamiltonian matrix leads to the emergence of one, and only one, energy level E_α below the lowest non-perturbed level.

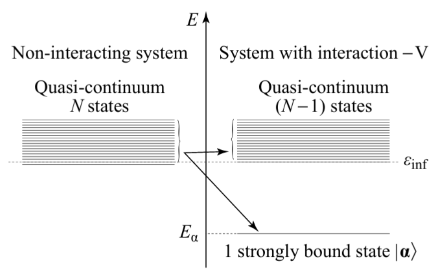

Figure 8.8 - System of ***N*** **levels forming a quasi-continuum**
With the interaction $-$ V, one, and only one, state emerges with energy below the lowest value of the continuum.

Figure 2: The system of N levels forms a quasi-continuum. With the interaction -V, one, and only one, the state emerges with energy below the lowest value of the continuum.

The eigenvector $|\psi\rangle$ associated with the eigenvalue E_α can be written in the basis of non-perturbed eigenvector as

$$|\psi\rangle = c_1|1\rangle + c_2|2\rangle + c_3|3\rangle + c_4|4\rangle$$

The c_i, components of the "bound" state $|\psi\rangle$ of the eigenvectors of the non-interacting Hamiltonian, satisfy the relations $c_1 > c_2 > c_3 > c_4$. Therefore, the states of lowest energy appear with a greater weight in the projection of $|\psi\rangle$. This means that in the dynamic picture described previously, according to which the electron jumps between the stationary states of the non-perturbed system with occupation probabilities c_j^2, the particle has a higher probability of being found in the states of lowest energy.

5 Two body interaction hamiltonian

When the particles interact with each other, we must add an interaction term $\hat{V}(r_1, r_2)$ to the two-particle Hamiltonian which becomes

$$\hat{H}_0(r_1, r_2) = -\frac{\hbar^2}{2m_1}\nabla_1^2 - \frac{\hbar^2}{2m_2}\nabla_2^2 + \hat{V}(r_1, r_2).$$

In this case, proceeding by separating variables is no longer possible. The search for new energy levels requires diagonalization of the matrix $\langle k_m k_n|\hat{V}|k_p k_s\rangle$ that, in the basis $|k_m k_n\rangle$ of eigenvectors of the non-interacting system of two particles, has in addition to the diagonal terms we have just seen, non-diagonal terms that are the matrix elements

$\langle k_p k_s|\hat{V}|k_m k_n\rangle = \frac{1}{V^2}\int V(r_1, r_2)e^{-i\mathbf{k}_p\cdot\mathbf{r}_1}e^{-i\mathbf{k}_s\cdot\mathbf{r}_2}e^{i\mathbf{k}_m\cdot\mathbf{r}_1}e^{i\mathbf{k}_n\cdot\mathbf{r}_2}d^3r_1d^3r_2.$

Integration over r_1 and r_2 extends over the box's volume. This matrix element takes the value in a unit volume, assuming the size of the box is much larger than the range of the potential (to avoid boundary effects) with $\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$:

$$\langle k_p k_s|\hat{V}|k_m k_n\rangle = \begin{cases} V(\mathbf{q}) & \text{if } \mathbf{q} = \mathbf{k}_s + \mathbf{k}_n - \mathbf{k}_m - \mathbf{k}_p, \\ 0 & \text{otherwise,} \end{cases}$$

where

$$V(\mathbf{q}) = \int V(\mathbf{r})e^{i\mathbf{q}\cdot\mathbf{r}}d^3r.$$

The process of interaction can be interpreted as follows :

- two electrons are initially in states $|m\rangle$ and $|n\rangle$ (momenta $\mathbf{k}_m$ and $\mathbf{k}_n$);
- during the interaction they exchange a momentum $\mathbf{q}$ with probability amplitude $V(\mathbf{q})$;
- After this exchange, the electrons are to be found in states $|p\rangle = |m - q\rangle$ and $|s\rangle = |n + q\rangle$ with momenta $(k_m - q)$ and $(k_n + q)$. Because this exchange must be without any change in total momentum, only terms satisfying the condition

  $$k_m + k_n = k_p + k_s$$

  are nonzero.

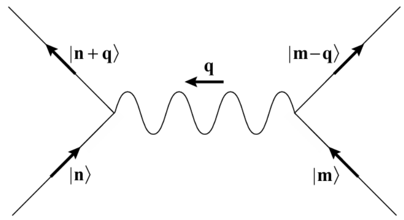

Figure 3: Diagrammatic presentation of the interaction between two particles In the initial state, the particles are in states $|m\rangle$ and $|n\rangle$. By exchanging momentum q, they are scattered into the states $|m-q\rangle$ and $|n+q\rangle$.

6 The BCS attractive interaction

The process is as follows:

1. an electron of initial momentum k_m "excites" a phonon of wave vector q transferring a momentum q; its final momentum is $(k_m - q)$;

2. the created phonon interacts with an electron of momentum k_n that "absorbs" it, acquiring a momentum $(k_n + q)$.

The electrostatic interaction between electrons and crystal ions causes the excitation and absorption of the phonon. The first electron (negative charge) displaces crystal ions (positive charge) and generates the phonon during its motion. The moving ions will interact with the second electron by giving up their movement. A detailed analysis reveals that to be non-vanishing, the non-diagonal elements of the interaction matrix should satisfy two conditions:

- The initial and final states must be states of opposite wave vectors and opposite spins. They are, therefore, of the form:

$$V_{k',k} = \langle k'_\uparrow, -k'_\downarrow | \hat{V}_{k,k} | k_\uparrow, -k_\downarrow \rangle$$

  . Such states $|k, k\rangle$ are called 'pair states'.

- As the particles exchanged during the interaction process are phonons, the only transitions possible are between pair states involving energy transfers less than the maximum phonon energy, which is order $E_D = \hbar\omega_D$ (the DEBYE energy 10 to 20 meV). As a consequence the only "pair states" that can be involved in the transitions are those constructed out of wave vectors of wavelength k whose energies k lie in a band of energies of order E_D around the FERMI energy E_F. The simple BCS approximation then considers that the non-zero matrix elements that describe the interaction are all equal with

$$V_{k'k} = \begin{cases} V & \text{if } E_F - E_D < (k \text{ and } k') < E_F + E_D \\ 0 & \text{otherwise} \end{cases}$$

7 Form for the actual attractive potential

The coupling $g_{\mathbf{q}}$ for longitudinal acoustic phonons in a jellium model (a simplified model of a metal) is related to the screened Coulomb interaction. It can be approximated as:

$$g_{\mathbf{q}} = i\sqrt{\frac{\hbar}{2MN\omega_{\mathbf{q}}}}qV_{\mathbf{q}},$$

where $V_{\mathbf{q}} = \frac{4\pi e^2}{q^2+k_s^2}$ is the screened Coulomb potential, k_s is the Thomas-Fermi screening wavevector, M is the ionic mass, N is the number of ions, and $\omega_{\mathbf{q}} = v_s q$ (for acoustic phonons, with v_s as the sound speed).

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7.1 Second-Order Perturbation Theory

The effective interaction between two electrons arises from the virtual exchange of a phonon, which we compute using second-order perturbation theory. The second-order energy correction to a state $|\psi_0\rangle$ (two electrons in states $(\mathbf{k}_1, \uparrow), (\mathbf{k}_2, \downarrow)$) is:

$$E^{(2)} = \sum_{n\neq 0} \frac{|\langle n|H_{\text{int}}|\psi_0\rangle|^2}{E_0 - E_n},$$

Where:

Initial state: $|\psi_0\rangle = c^\dagger_{\mathbf{k}_1\uparrow}c^\dagger_{\mathbf{k}_2\downarrow}|0\rangle$, with energy $E_0 = \epsilon_{\mathbf{k}_1} + \epsilon_{\mathbf{k}_2}$,

Intermediate state $|n\rangle$: One electron scatters by emitting a phonon, e.g., electron 1 from $\mathbf{k}_1$ to $\mathbf{k}_1 + \mathbf{q}$, producing a phonon $\mathbf{q}$, then electron 2 absorbs it, scattering from $\mathbf{k}_2$ to $\mathbf{k}_2 - \mathbf{q}$.

The intermediate state is:

$$|n\rangle = c^\dagger_{\mathbf{k}_1+\mathbf{q},\uparrow}c^\dagger_{\mathbf{k}_2-\mathbf{q},\downarrow}a^\dagger_{\mathbf{q}}|0\rangle,$$

where $c^\dagger_{\mathbf{k}}$ represents electron creation operators and $a^\dagger_{\mathbf{q}}$ represents phonon creation operators, with energy:

$$E_n = \epsilon_{\mathbf{k}_1+\mathbf{q}} + \epsilon_{\mathbf{k}_2-\mathbf{q}} + \hbar\omega_{\mathbf{q}}.$$

The energy difference is:

$$E_0 - E_n = (\epsilon_{\mathbf{k}_1} + \epsilon_{\mathbf{k}_2}) - (\epsilon_{\mathbf{k}_1+\mathbf{q}} + \epsilon_{\mathbf{k}_2-\mathbf{q}} + \hbar\omega_{\mathbf{q}}).$$

7.2 Matrix Elements

The matrix element for the emission process is:

$$\langle n|H_{\text{int}}|\psi_0\rangle = g_{\mathbf{q}},$$

Since H_{int} connects the initial state to the intermediate state by creating a phonon and scattering an electron. The second step (absorption) leads to a final state, but in the effective potential, we sum over intermediate states.

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7.3 Effective Interaction Potential

The effective interaction potential V_{eff} between two electrons is obtained by interpreting the second-order energy shift as an interaction term. For electrons scattering from $(\mathbf{k}_1, \mathbf{k}_2)$ to $(\mathbf{k}_1', \mathbf{k}_2')$, with momentum transfer $\mathbf{q} = \mathbf{k}_1' - \mathbf{k}_1 = \mathbf{k}_2 - \mathbf{k}_2'$, the effective potential is:

$$V_{\text{eff}}(\mathbf{q}, \omega) = \sum_{\text{intermediate states}} \frac{|\langle \text{final}|H_{\text{int}}|n\rangle\langle n|H_{\text{int}}|\text{initial}\rangle|}{E_{\text{initial}} - E_n},$$

where ω is the energy difference between initial and final states in frequency units ($\hbar\omega = \epsilon_{\mathbf{k}_1} + \epsilon_{\mathbf{k}_2} - \epsilon_{\mathbf{k}_1'} - \epsilon_{\mathbf{k}_2'}$).

The matrix elements give $|g_{\mathbf{q}}|^2$, and the energy denominator, for Cooper pairs near the Fermi surface ($\epsilon_{\mathbf{k}_1} \approx \epsilon_{\mathbf{k}_2} \approx E_F$), simplifies. The energy difference in the denominator becomes dominated by the phonon energy if the electron energy difference is small:

$$\epsilon_{\mathbf{k}_1} + \epsilon_{\mathbf{k}_2} - \epsilon_{\mathbf{k}_1+\mathbf{q}} - \epsilon_{\mathbf{k}_2-\mathbf{q}} \approx 0,$$

so:

$$E_0 - E_n \approx -\hbar\omega_{\mathbf{q}}.$$

However, including the frequency dependence ($\hbar\omega$) of the scattering process, the denominator is:

$$\hbar\omega - \hbar\omega_{\mathbf{q}}.$$

Thus, the effective potential from phonon exchange is:

$$V_{\text{eff}}(\mathbf{q}, \omega) = \frac{|g_{\mathbf{q}}|}{\hbar\omega - \hbar\omega_{\mathbf{q}}}.$$

For the reverse process (phonon absorption then emission), the denominator becomes $\hbar\omega + \hbar\omega_{\mathbf{q}}$, so the total contribution is:

$$V_{\text{eff}}(\mathbf{q}, \omega) = |g_{\mathbf{q}}| \left(\frac{1}{\hbar\omega - \hbar\omega_{\mathbf{q}}} + \frac{1}{\hbar\omega + \hbar\omega_{\mathbf{q}}} \right).$$

Simplify:

$$\frac{1}{\hbar\omega - \hbar\omega_{\mathbf{q}}} + \frac{1}{\hbar\omega + \hbar\omega_{\mathbf{q}}} = \frac{2\hbar\omega}{(\hbar\omega)^2 - (\hbar\omega_{\mathbf{q}})^2} = \frac{2\omega}{\omega^2 - \omega_{\mathbf{q}}^2}.$$

Thus:

$$V_{\text{eff}}(\mathbf{q},\omega) = |g_{\mathbf{q}}| \frac{2\omega}{\omega^2 - \omega_{\mathbf{q}}^2}.$$

— Computing $|g_{\mathbf{q}}|$ Using the expression for $g_{\mathbf{q}}$:

$$g_{\mathbf{q}} = i\sqrt{\frac{\hbar q^2}{2MN\omega_{\mathbf{q}}} \frac{4\pi e^2}{q^2 + k_s^2}},$$

$$|g_{\mathbf{q}}| = \left(\frac{\hbar}{2MN\omega_{\mathbf{q}}}\right) q^2 \left(\frac{4\pi e^2}{q^2 + k_s^2}\right).$$

For acoustic phonons, $\omega_{\mathbf{q}} = v_s q$, and $MN = \rho V$ (where ρ is the mass density, V is the volume). However, in the jellium model, we approximate the prefactor. The key term is:

$$|g_{\mathbf{q}}| \propto \frac{q^2}{\omega_{\mathbf{q}}} \left(\frac{4\pi e^2}{q^2 + k_s^2}\right).$$

The canonical result, after averaging over all electron modes, and approximating $\omega \approx \omega_D$ gives:

$$|g_{\mathbf{q}}| \approx \frac{\hbar\omega_{\mathbf{q}}^2}{2} \left(\frac{4\pi e^2}{q^2 + k_s^2}\right) \frac{1}{\hbar\omega} = \frac{\omega_{\mathbf{q}}^2}{2\omega} \left(\frac{4\pi e^2}{q^2 + k_s^2}\right).$$

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7.4 Final Form of the Second Term

Substitute $|g_{\mathbf{q}}|$:

$$V_{\text{eff,perturbation}}(\mathbf{q},\omega) = \frac{1}{2} \left(\frac{4\pi e^2}{q^2 + k_s^2}\right) \frac{2\omega_q}{\omega^2 - \omega_{\mathbf{q}}^2}.$$

The total potential is adding the screening term:

$$V_{\text{eff,total}}(\mathbf{q},\omega) = \left(\frac{4\pi e^2}{q^2 + k_s^2}\right) + \frac{4\pi e^2}{q^2 + k_s^2} \frac{\omega_{\mathbf{q}}^2}{\omega^2 - \omega_{\mathbf{q}}^2}.$$

This is attractive for $\omega_{\mathbf{q}} > \omega$.

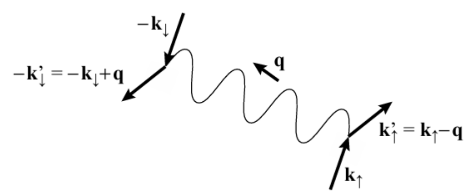

Figure 4: Graph of the interaction at the origin of BCS superconductivity Initially, two electrons occupy the pair state $|k, -k\rangle$. At time t, the electron of wave vector k creates a phonon of wave vector q and acquires the wave vector $k' = k - q$. At a later time, the electron of wave vector $-k$ absorbs the phonon and its wave vector becomes $k' = -k + q$. The net effect is that the electron pair has been scattered from the state $|k, -k\rangle$ into the state $|k', -k'\rangle$.